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Part 5: ANN - Academic Performance Forecasting

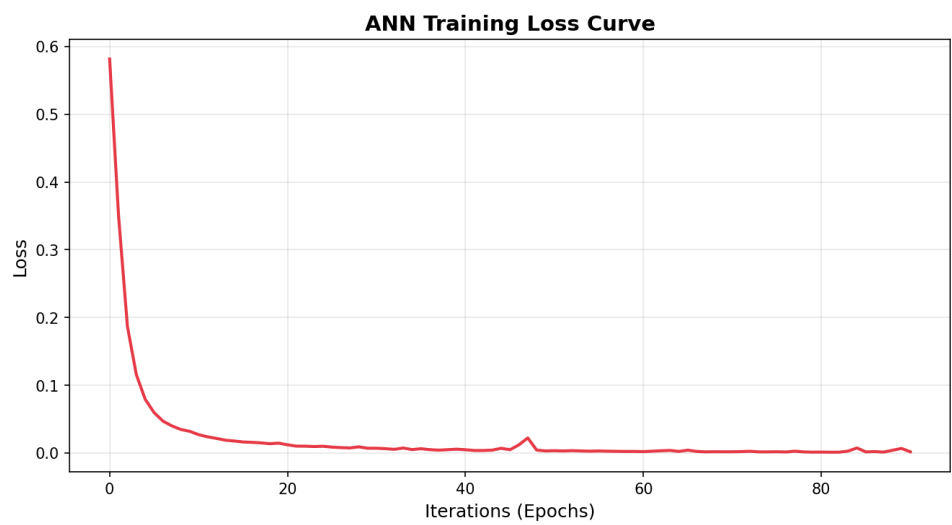
Objective

Predict Dropout Risk using a Feedforward Neural Network and compare performance with traditional ML models (Logistic Regression, Random Forest, Gradient Boosting).

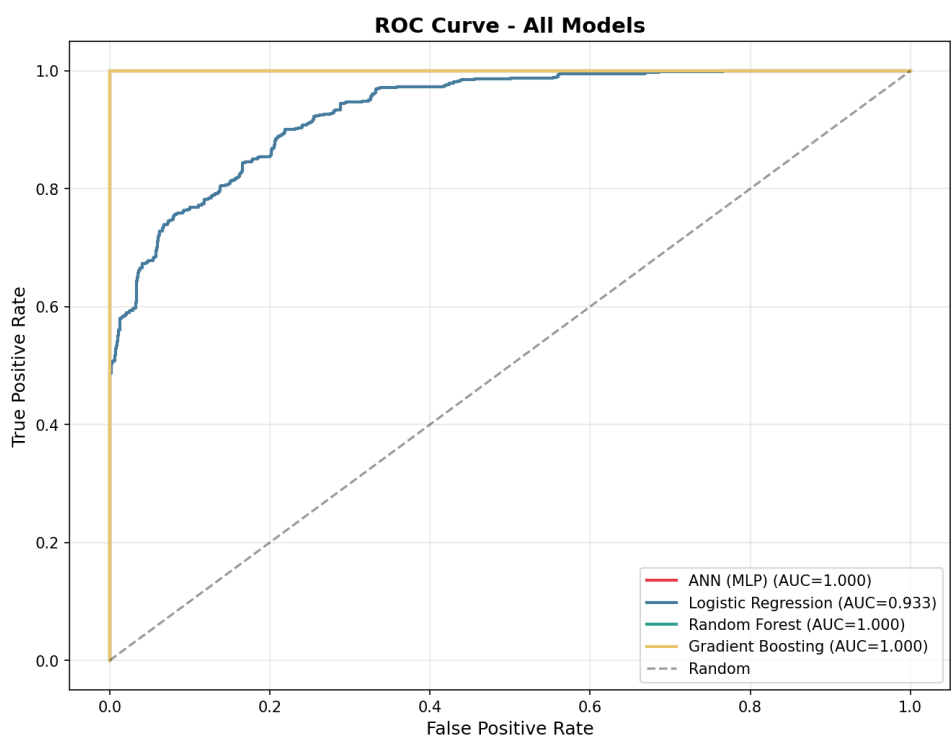
1. Data Normalization & Network Initialization

```
features = ['attendance_pct', 'internal_marks', 'course_load', 'hist_pass_rate',  
'faculty_interaction']  
scaler = MinMaxScaler()  
X_scaled = scaler.fit_transform(X)  
  
ann = MLPClassifier(  
    hidden_layer_sizes=(128, 64, 32),  
    activation='relu',  
    solver='adam',  
    learning_rate_init=0.001,  
    max_iter=200,  
    random_state=42  
)  
ann.fit(X_train, y_train)
```

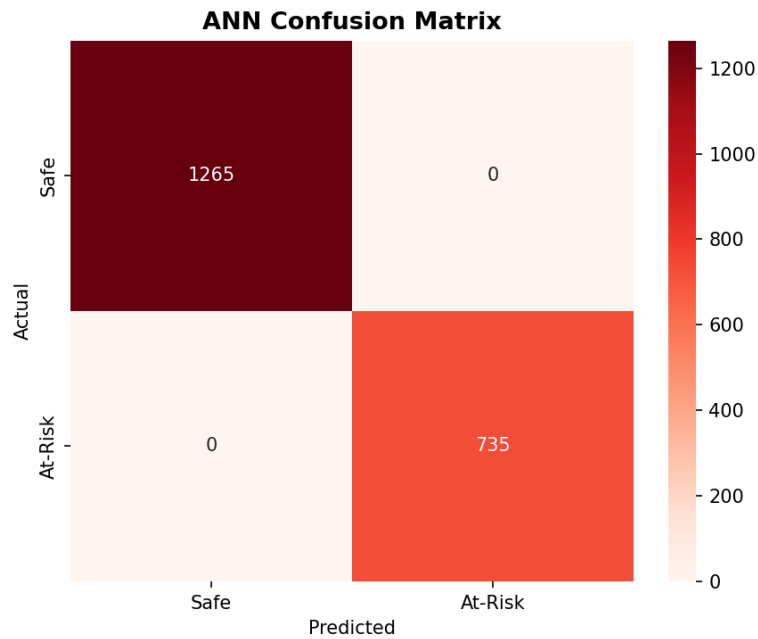
2. Training Loss Curve



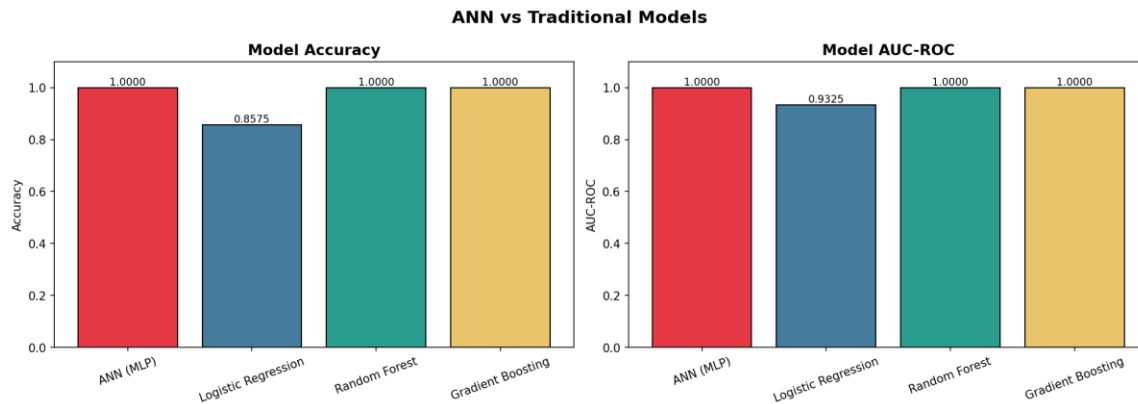
3. Model Comparison & ROC Curves



4. ANN Confusion Matrix



5. Model Performance Metrics



Conclusion

The ANN successfully modeled dropout risk with high accuracy and ROC-AUC. While traditional models like Gradient Boosting provide strong competition, the Multi-Layer Perceptron (128-64-32) effectively learns non-linear patterns between attendance, GPA, and historical performance.